

Terms

$d ::= \epsilon$ for $e \in \mathbb{R}$

$| d \text{ op } d$ op $\in \{ +, -, \cdot, / \}$

$t ::= d$
 $| \text{coin}()$ $| \text{rand}()$
 $| \text{if } \text{opt}$

$b ::= \text{true} \mid \text{false}$

$| d \bowtie d$ $\bowtie \in \{ =, <, ? \}$
 $| \neg d$

Programs

$e ::= \text{skip}$

$| x := e$

$| e ; e$

$| \text{if } b \text{ then } e \text{ else } e$

$| \text{while } b. \text{ do } e$

Operational Semantics

$s \xrightarrow{e} s'$

$\llbracket t \rrbracket : \mathbb{R}^n \times [0, 1]^\omega \times \mathbb{B}^\omega \rightarrow \mathbb{R} \times [0, 1] \times \mathbb{B}^\omega$

$\llbracket v \rrbracket = \overline{(\underline{s}, m, p)} \mapsto (v, m, p)$

$$\llbracket x_i \rrbracket : (s, m, p) \mapsto (s_i, m, p)$$

$$\llbracket \text{coin}() \rrbracket : (s, m, p) \mapsto (\text{hd } p, m, \text{tl } p)$$

$$\llbracket \text{rand}() \rrbracket : (s, m, p) \mapsto (\text{hd } m, \text{tl } m, p)$$

$$\llbracket t_1 \text{ op } t_2 \rrbracket : (s, m, p) \mapsto$$

$$\searrow \text{let } (s_1, m_1, p_1) = \llbracket t_1 \rrbracket (s, m, p)$$

$$\text{let } (s_2, m_2, p_2) = \llbracket t_2 \rrbracket (s_1, m_1, p_1)$$

$$(s_2 \text{ op } s_2, m_2, p_2)$$

$$\llbracket b \rrbracket : \mathbb{R}^n \times [0, 1]^w \times \mathbb{B}^w \rightarrow \mathbb{B}$$

$$\llbracket t_1 \stackrel{?}{=} t_2 \rrbracket : (s, m, p) \mapsto \begin{cases} \text{true} & \text{if } \llbracket b \rrbracket (s, m, p) = \llbracket t_2 \rrbracket (s, m, p) \\ \text{false} & \text{otherwise} \end{cases}$$

$$\llbracket b_1 \& b_2 \rrbracket : (s, m, p) \mapsto$$

$$\llbracket b_1 \rrbracket (s, m, p) \wedge \llbracket b_2 \rrbracket (s, m, p)$$

$$\llbracket !b \rrbracket : (s, m, p) \mapsto \neg (s, m, p)$$

$$(e, s, m, p) \longrightarrow (e', s', m', p')$$

$$(e, s, m, p) \longrightarrow^{\Delta} (\text{skip}, s', m', p')$$

$$\llbracket t \rrbracket (s, m, p) = (e, m', p')$$

$$(x_i := t, s, m, p) \rightarrow (\text{skip}, s[x \mapsto t], m, p)$$

$$\frac{(e, s, m, p) \rightarrow (e', s', m', p')}{(e; e_2, s, m, p) \rightarrow (e'; e_2; s', m', p')}$$

$$(\text{skip}; e, s, m, p) \rightarrow (e, s, m, p)$$

$$\llbracket b \rrbracket (s, m, p) = \text{tt}$$

$$(\text{if } b \text{ then } e_1 \text{ else } e_2, s, m, p) \rightarrow (e_1, s, m, p)$$

$$\llbracket b \rrbracket (s, m, p) = \text{ff}$$

$$\quad \rightarrow (e_2, s, m, p)$$

$$(\text{while } b \text{ do } e, s, m, p)$$

$$\rightarrow (\text{if } b \text{ then } e; \text{while } b \text{ do } e \text{ else skip}, s, m, p)$$

$$(e, s, m, p) \xrightarrow{*} (e, s, m, p)$$

$$(e, s, m, p) \rightarrow (e', s', m', p')$$

$$(e, s, m, p) \rightarrow^* (e', s', m', p')$$

$$(e \text{ skip } p) \rightarrow e' \text{ skip } m' p' \quad . \quad - \quad -$$

$$\{ \text{while } x=0 \text{ do } x := \text{coin}() \text{ ; } 0, \underline{0000} \dots \}$$

$$\rightarrow^* (\text{skip}, 1, \dots)$$

Random walk on $\mathbb{Z}^2$

$$(step, s, \overset{\sim}{00} m, p) \rightarrow^* (\text{skip}, s[(u,v) \mapsto (u,v-1)] \\ (x,y) \mapsto (0,0), m, p)$$

$$0 \mid m \rightarrow^* \begin{array}{l} (u,v) \mapsto (u-1,v) \\ (x,y) \mapsto (0,1) \end{array}$$

$$1 \mid 0 \mid m \rightarrow^* (\text{skip}, s[(u,v) \mapsto (u+1,v)] \\ (x,y) \mapsto (1,0)]$$

$$1 \mid m \rightarrow^* \begin{array}{l} s[(u,v) \mapsto (u,v+1)] \\ (x,y) \mapsto (1,1)] \end{array}$$

Step (u, v)

$$x := \text{coin}(C)$$

$$y := \text{coin}(C)$$

$$u := u + (x - y)$$

$$\hookrightarrow v := v + (x + y - 1)$$

$$\llbracket e \rrbracket : M\mathbb{R}^n \rightarrow M\mathbb{R}^n$$

state transformer

$$\llbracket x := \text{coin}(C) \rrbracket s \rightarrow \frac{1}{2} s[x \mapsto 0] + \frac{1}{2} s[x \mapsto 1]$$

$$\llbracket r \rrbracket (e_1, \dots, e_n) = \delta_r, \quad r \in \mathbb{R}$$

$$\llbracket x_i \rrbracket (\dots e_i \dots) = \delta_{e_i}$$

$$\llbracket \text{coin}(C) \rrbracket (\dots) = \frac{1}{2} \delta_0 + \frac{1}{2} \delta_1$$

$$\llbracket \text{rand}(C) \rrbracket (\dots) = \lambda$$

$$\llbracket t_1 \text{ op } t_2 \rrbracket (\vec{e}) = \text{op}_\# (\llbracket t_1 \rrbracket (\vec{e}) \otimes \llbracket t_2 \rrbracket (\vec{e}))$$

push forward

$$\llbracket t_1 = t_2 \rrbracket = \lambda(\vec{e}) (\llbracket t_1 \rrbracket (\vec{e}) = \llbracket t_2 \rrbracket (\vec{e}))$$

...

$$\text{maps: } M\mathbb{R}^n \rightarrow M\mathbb{R}^n \leftarrow$$

$$[\text{skip}] = \text{id}_{M\mathbb{R}^n} : M\mathbb{R}^n \rightarrow M\mathbb{R}^n$$

$$[t] : \mathbb{R}^n \rightarrow M\mathbb{R}$$

$$F_t^i : \mathbb{R}^n \rightarrow M\mathbb{R}^n$$

$$F_t^i(x_1, \dots, x_n) = \delta_{x_1} \otimes \dots \otimes \delta_{x_{i-1}} \otimes \\ [t](x_1, \dots, x_n) \otimes \delta_{x_{i+1}} \otimes \dots \\ \otimes \delta_{x_n}$$

$$[x_i : t](\mu) = (F_t^i)_* (\mu)$$

$$[x_i := r](\mu)(B_1, \dots, B_n)$$

$$= \int \delta_{x_1} \otimes \dots \otimes \delta_r \otimes \dots \otimes \delta_{x_n} \\ (B_1 \otimes \dots \otimes B_n) d\mu$$

$$[x_i = \text{coinC}](\mu)(B_1, \dots, B_n)$$

$$= \mu(B_1 \times \dots \times \mathbb{R} \times \dots \times B_n) -$$

$$\left(\frac{1}{2}\delta_0 + \frac{1}{2}\delta_1\right)(B_i)$$

$$\llbracket e_1 ; e_2 \rrbracket = \llbracket e_1 \rrbracket \circ \llbracket e_2 \rrbracket$$

$$\llbracket \text{if } b \text{ then } e_1 \text{ else } e_2 \rrbracket = \\ \llbracket e_1 \rrbracket \top \llbracket b \rrbracket + \llbracket e_2 \rrbracket \top \llbracket b \rrbracket^c$$

$$\llbracket \text{while } b \text{ do } e \rrbracket = \llbracket \text{while } b \text{ do } e \rrbracket \circ \llbracket e \rrbracket \cdot \top \llbracket b \rrbracket \\ + \top \llbracket b \rrbracket^c$$

$$\tau(S) = S \circ \llbracket e \rrbracket \cdot \top \llbracket b \rrbracket + \top \llbracket b \rrbracket^c$$

$$\llbracket \text{while } b \text{ do } e \rrbracket \\ = \bigvee_{n \geq 0} \tau^n(e)$$

- monotone
- bounded $\rightarrow$ LFP